

A Parallel Duplicate Address Detection (PDAD) Mechanism to Reduce Handoff Latency of Mobile Internet Protocol Version 6 (MIPv6)

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Abstract— Next Generation Networks (NGN) are envisioned to be fully Internet Protocol (IP) based so that users can experience high quality and ubiquitous communication. In order to realize this, Internet Engineering Task Force (IETF) is taking advantage of Mobile Internet Protocol version 6 (MIPv6) because it has more suitable features than MIPv4. However, MIPv6 is not widely deployed due to handoff latency and other limitations. This causes packet loss and degrades the Quality of Service (QoS) for real-time applications. MIPv6 handoff latency includes link layer establishment delay, movement delay, address auto-configuration delay and binding update/registration delay. Duplicate Address Detection (DAD) is performed during address auto-configuration which takes around 70% of the time for the total handoff procedures. In order to reduce this handoff latency, a Parallel DAD (PDAD) model is proposed for reducing packet loss. The key idea behind this model is that a Mobile Node (MN) solicits for a router advertisement (RA) in parallel to its neighbor routers before leaving the previous point of attachment.

Keywords- Mobile IPv6 (MIPv6), Duplicate Address Detection (DAD), Parallel DAD (PDAD) Handoff Latency, Seamless Communcation.

I. INTRODUCTION

Wireless communication is a fast growing area due to the increasing number of users and their growing demand for better quality of service. However, these demands becomes challenging because seamless networks are needed. In a seamless network, mobile nodes move freely from one cell to another without disconnecting. Seamless communication is important for network-enabled applications to operate continuously at the desired quality of service in a wired or wireless IP network, especially for real time applications such as audio and video streaming. In addition to this, it reduces packet loss and improves Quality of Services (QoS). Several strategies have been proffered as solutions however MIPv6 provides better solutions due to its wide address spaces. When mobile users migrate from the coverage of one network access point to another, handoff latency is introduced which is one of the major causes of performance degradation. Whenever a mobile node (MN) moves from its registered home agent (HA) to another area, it needs to be temporarily registered with that

agent by getting a care-of-address (CoA). With this CoA, MN maintains communication with its HA and other corresponding nodes (CN) that wants to communicate with the MN. To make a mobile node (MN) stay connected to the Internet regardless of its location, MIPv6 [1] is proposed as the next generation wireless Internet protocol. This is achieved by using the CoA to indicate the location of the MN. Although MIPv6 has many promising features and presents an elegant mechanism to support mobility, it has an inherent drawback in the handover process where there is a short period that the MN is unable to send or receive packets due to link switching delay and IP protocol operations [2]. Handover is the process by which an MN keeps its connection active when it moves from one access medium to another [3]. The handoff process should accomplish four operations [4]: link layer establishment, movement detection, new CoA configuration, and binding update (BU). During the handoff period, the MN is unable to send or receive packets. The length of this period which is called handoff latency is very critical for delay-sensitive and real-time services [5]. To improve the performance of MIPv6, there are many improvements for enhancing the handoff process. The aim of these enhancements is to reduce the latency and the number of packet loss during the handoff period.

In this paper we propose a mechanism to reduce handoff latency especially for duplicate address detection (DAD). According to our proposed model, the MN requests to generate and check uniqueness of a CoA to its neighbor routers in parallel. Each requested router generates a CoA for the respected MN and checks the uniqueness of that address in advance. Whenever the MN moves from its HA to any visited networks it may experience with handoff for the first time but for subsequent transition there is no need to perform DAD algorithm by the MN as CoA has already been checked in advance. The details are described in section V.

The rest of this paper is organized as follows. In section II, we discuss the overview of MIPv6 in a brief with its facilities and functionalities. In section III, MIPv6 delays are reviewed. Hierarchical MIPv6 (HMIPv6), Fast MIPv6, Optimistic DAD and Advance DAD (ADAD) are discussed in section IV. In section V, we described our proposed mechanism Parallel

Duplicate Address Detection (PDAD) model. Simulation work is discussed in section VI. Section VII concludes the paper.

II. OVERVIEW OF MOBILE IPV6

MIPv6 [1] has been proposed to accommodate the increasing demand of mobility in the Internet by IETF. Based on the proposal, an MN should generate a permanent and temporary IP address for further communication [6]. The permanent address is assigned by home agent (HA) to the MN as a global identifier that is fixed. A temporary address also known as care-of address (CoA) is obtained from the visiting network which indicates host's actual location that changes with time as MN moves to foreign networks. While an MN is attached to its HA, it is able to receive packets destined to its HA by existing routing mechanism. However, whenever it visits any foreign networks, the home address will be invalid. In that case, after obtaining CoA, the MN sends a binding update (BU) to its HA and HA maps the current CoA in its binding cache. To keep this cache updated, each MN sends a BU message containing current CoA information to its HA. The MN sends this BU periodically or whenever it changes its current location [7]. In the meantime, the corresponding nodes (CN) wishing to communicate with the MN also maintain a binding cache. If the CN does not have an active map address for a mobile node in its binding cache, it will send the packets to MN's previous home address for the first time. The HA then encapsulates packets and tunnels them to the MN's CoA. Thus, MN can communicate with CN directly and avoid a triangular path.

CoA plays an important role for seamless communication as MN moves within subnets. To get this CoA, MIPv6 uses stateful or stateless auto-configuration [8]. MN should run duplicate address detection (DAD) algorithm before assigning the address to its interface to verify the uniqueness of the CoA. The algorithm confirms the availability of a unique CoA. It is mandatory for MN to complete DAD procedure before registering the CoA to its HA and CN by using BU.

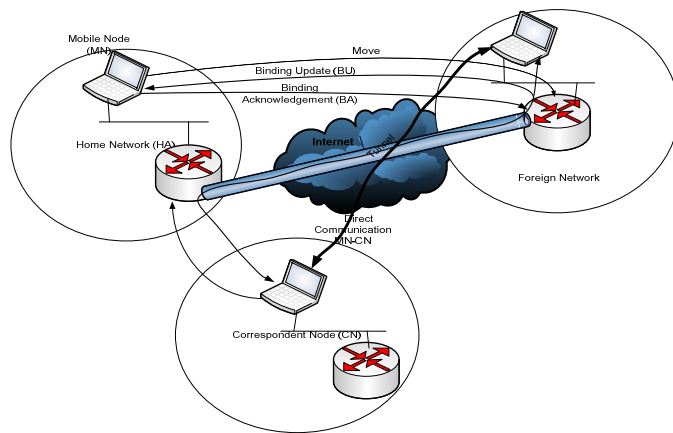


Figure 1. Standard MIPv6

In IPv6 network, the probability of address conflict is usually very low [9] and it is independent of stateless or

stateful auto-configuration. In case of stationary MN, the address conflict might not be a problem. However, in case MN moves among networks and tries to maintain its current session, address conflict can pose big problem. The session between the MN and CN may be broken and the rightful owner of that address may receive unexpected packets. Therefore, RFC 2462 [8] and RFC 3775 [1] proposed that DAD should still be performed on all new CoAs.

Moreover, IEEE designed a 64-bit Extended Universal Identifier (EUI-64) to attain global uniqueness but this could not be maintained because manufacturers use unregistered 802.x addresses and also due to random change of MAC address by users or obscure network devices [10]. Furthermore, some vendors are shipping batches of cards with the same MAC address. Therefore, DAD process is essential for detecting and handling duplicate addresses.

III. DELAYS OF MIPv6

A. Identifying handoff latency components

Whenever the received signal strength indicator (RSSI) of the MN is decreasing, it means the MN is going to perform handoff [5]. The handoff is disconnected from the network connection to stop receiving packets from CN until MN moves to a new subnet and receives packets from CN again. To initiate handoff process, when the MN moves to a new subnet, it sends router solicitation (RS) to the neighbor routers as router advertisement (RA). Based on the information of RA and MN's MAC address, MN creates CoA. According to IPv6 specifications, the MN needs to be sure that its link-local address in the new link is unique by implementing the DAD procedure. After completing the DAD process, the auto-configuration procedure is accomplished by the MN to form its new CoA. To check this CoA's uniqueness, another DAD procedure is conducted. This takes time for the overall handover process and degrades the QoS. After obtaining a CoA, the MN registers a BU to its HA and CN. In this way, the handoff process is completed and packets from the CN can be delivered to the MN. The handover procedure in MIPv6 consists of two types [11], link layer (L2) and network layer (L3). L2 handoff is a process by which an MN changes its physical link-layer connection to another when it moves from an old access point (AP) to a new AP. In L3 handoff, a MN identifies that it has moved to a new link layer where new subnet prefix is used. These MNs will change their primary CoA to new one. As MN moves, change of AP followed by the change of the subnet leads to L3 handoff. L3 handoff can be divided into three types [4], movement detection, configuration including DAD for CoA, and binding update. Each of the three components causes non-negligible delays where configuration delay takes longer time.

1) **Link Layer Establishment Delay:** The time required by the physical interface to establish a new association between old and new AP.

2) **Movement Detection Delay:** The duration between sending RS to the new AP and receiving RA from new AP when the MN moves to a new network. Alternatively, the MN

has to wait for RA as it broadcasts to the new MN for a minimum of 30ms to a maximum of 70ms.

3) **Configuration or DAD Delay:** The duration an MN receives an RA to the time an MN is configured with a new CoA based on the information of the RA and MN's MAC address. It also includes time for performing DAD for CoA.

4) **BU or Registration Delay:** The duration an MN sends a BU to its HA and CN to the time it receives an acknowledgement and first data packets to the MN respectively. In MIPv6, the binding update (BA) from CN is not essential and it is assumed that handoff process is completed if packets have been received by MN.

B. Analysis of Delays

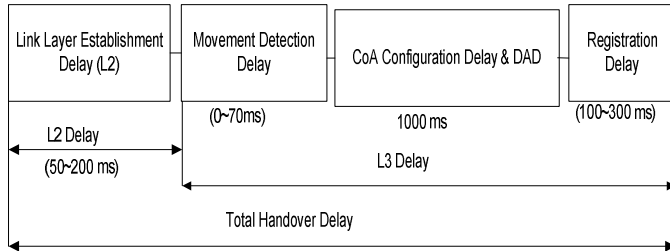


Figure 2. Handoff delays of Mobile IPv6

Let the link layer establishment delay be denoted as D_L , movement detection delay as D_M , address configuration delay as D_C , and registration delay as D_R and overall delay can be denoted as D_{MIPv6} that is:

$$D_{MIPv6} = D_L + D_M + D_C + D_R$$

To minimize the handoff latency of standard MIPv6, these delay components should be mitigated. The L2 delay is the shortest time among all delays. The value considered in [12, 13] is between 50ms to few hundred milliseconds. This delay is dependent on physical medium and changes from device to device. This delay can be minimized by the scanning, authenticating and re-associating phase. The movement detection delay component can be optimized by decreasing the router advertisement interval or by using a link layer trigger so that the MN always actively request for the router advertisement by sending a router solicitation. However, the configuration of the CoA and registration delay is the longest part of handoff latency. According to [19], the CoA configuration with DAD takes 1000 ms and binding update registration procedures needs around 100-300 ms and the total time required is more than 15000 ms according to simulations as described in Fig. 2. However, in real network it may vary from place to place and time to time. In this paper, we only focus on configuring CoA that includes DAD. This is essential for the address configuration process.

IV. RELATED WORKS

A. Hierarchical MIPv6 (HMIPv6)

MIPv6 faces many problems related to handoff latencies resulting in high packet loss rate and degradation of QoS. To minimize handoff latencies, several mechanisms have been proposed. MIPv6 sends BU to its HA and CN whenever it changes to a new subnet. However, HMIPv6 [14] reduces these signaling overhead by introducing a mobility anchor point (MAP) that separates mobility management into intra-domain mobility and inter-domain mobility. When an MN enters a MAP domain, it is given two addresses, regional address (RCoA) and on-link address (LCoA). The MN then sends a BU to the MAP that binds RCoA and LCoA. MN also registers its RCoA with the HA and CN. If the MN moves within the same MAP domain, it only changes its LCoA and updates the HA and CN. On the other hand, if it changes the MAP domain then it also needs to configure LCoA and RCoA and sends a BU to the HA and CN. However, MAP failure makes the whole communication disconnected. Service and performance is not considered while HMIPv6 is designed.

B. Fast MIPv6 (FMIPv6)

In FMIPv6 [15], partial handover of L3 is performed in advance by anticipating the movement of the MN to a new subnet. A tunnel is established between a currently attached access router and an anticipated router in order not to lose any packets during the handover process. FMIPv6 allows the MN to quickly detect the movement of an MN for configuring a CoA by using new access point's information. The mobile node starts fast handover when an L2 trigger takes places. The MN sends a Router Solicitation for Proxy Advertisement (RtSolPr) message to its current access router for a Proxy Router Advertisement (PrRtAdv) message. The MN forms a new CoA and sends a BU message based on the information provided in the Proxy Router Advertisement message. However, this protocol is developed based on anticipation that does not guarantee reliable packet transmission. The exact time of MN's movement is uncertain and unpredictable which may lead to packet loss.

C. Optimistic DAD

The optimistic DAD [16] method is developed based on the assumption that the probability of a 64 bit address conflict is very small because the address is randomly generated. The advantage of optimistic DAD is that it can cooperate with standard DAD without adding any new device on the routers. Moreover, it maintains the stateless characteristic of IPv6. Without checking any DAD procedure, optimistic DAD allows MN to use a tentative address from router's cache. After configuring an address, if the duplicate address is detected by other node, the MN must immediately reconfigure that address. This process degrades the quality of previous node and current MN. Normally, the duplication of address depends on the capacity and size of the network. Large networks with many users may experience address collision. As a matter of fact, it is difficult to appropriately configure a large network. This means collision is not negligible. If collision occurs, the recovery time of optimistic DAD could be even longer than standard DAD.

D. Advanced DAD

Advance DAD (ADAD) [17,18] allows the access routers (AP) to store unique address in its pool after generating a random address and performing DAD. ADAD requires exchange of modified RS and RA messages to help MN obtain duplicate free address from AR. The AP listens to other nodes by using neighbor discovery messages in the network. If another node is using the same address in its pool, it must remove that address from its cache, generate and test a new address. Whenever an MN moves to the new AP's region it newly configures an address without further delay. Furthermore, AR must hide neighbor advertisement messages using duplicate address which is free to any nodes.

V. PROPOSED MECHANISM

Duplicate address detection (DAD) time is the longer phase of handoff transition from current AR to visiting AR. The goal of our mechanism is to configure a CoA in advance in a parallel fashion by accessing neighbor routers. The new CoA is generated by multiple ARs using a combination of MN's L2 address and subnet prefix information and then DAD is performed. This mechanism is applicable to common MIPv6 networks and reactive model of FMIPv6 networks. The simple mechanism is after noticing the RSSI value is decreasing and the MN moves to a new subnet, it sends RS to the new routers containing MN's L2 address to all neighbor routers. Here the MN needs to solicit to all neighbour routers. The neighbor routers will form a CoA address by combination of MN's L2 address and router's subnet information and stores this address in its cache for the requested MN. The individual router checks the duplication of the newly generated CoA in the same network. Whenever an MN moves to a new subnet, it configures the stored address without performing DAD since it is checked earlier. Each router keeps this CoA for a specific time in its cache as MN may move to any other neighboring routers so that the MN can easily configure previously generated CoA. If an address collision occurs, the router needs to generate another address and perform DAD on that address. Whenever an MN moves further to a neighboring router, it can be configured by a reply message to the MN. After a specific time the routers release the stored address from its cache and assumes that the MN changed its movement to other direction. If the MN proceeds to a fully new routers area then it needs to follow the same procedures as stated above.

For example, in Fig. 3 the MN is registered with its HA in the middle of the network. Whenever the MN intends to move from HA to foreign networks, it asks for CoA to the neighboring foreign networks by requesting RS in parallel or simultaneously. Here, the foreign networks 1, 2 and 3 generate 3 CoAs individually and store it in their cache after checking the uniqueness of each address. Moreover, when the MN moves from foreign network 1 to 2 or 2 to 3, it does not need to request for any address generation. From Fig. 4, it shows the signaling flow of the total network. Based on the RSSI from the old AR, the MN detects it is going to change its active point of attachment and about to perform handoff.

Meanwhile the MN gets new signals from AR1, AR2 and AR3 and sends RS to these new routers. The 3 routers generate 3 CoA and send neighbor solicitation (NS) messages to all neighboring routers and MN. After checking and validating the uniqueness of that address among the routers and other MN or CN in that network, the respected router stores that unique address in its cache for a time period. The cached address can be configured whenever the MN moves to previously requested router's area. The MN then sends BU message to the HA and CN to connect for data transmission.

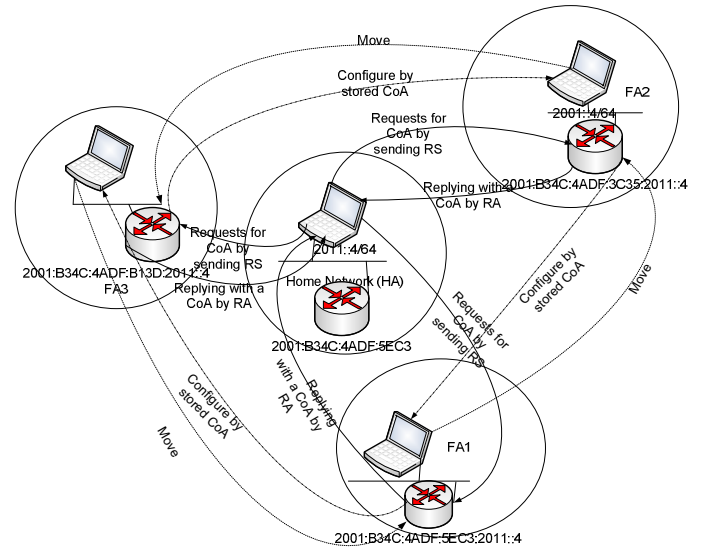


Figure 3. Proposed Mechanism

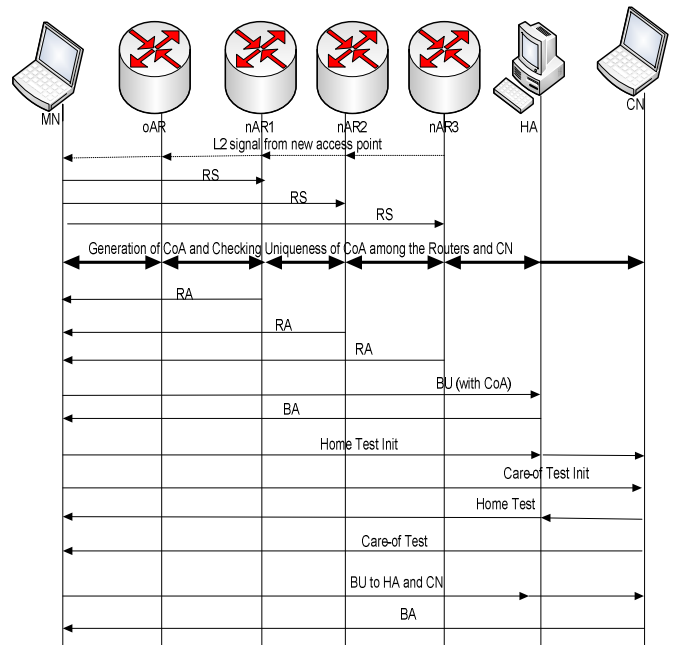


Figure 4. Signal Flow of PDAD Model

VI. SIMULATION SCENARIO COMPARISON

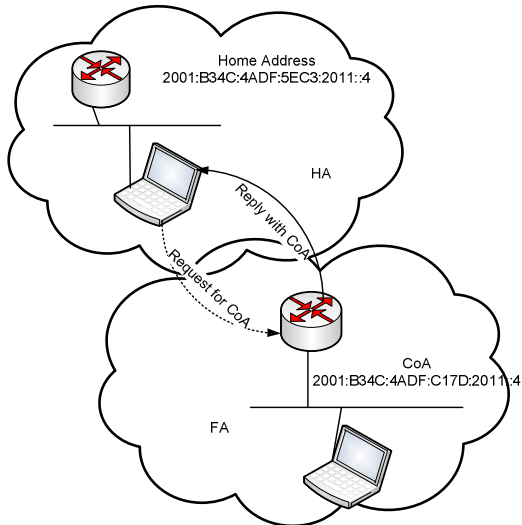


Figure 5 (a). Existing MIPv6 address configuration scenario

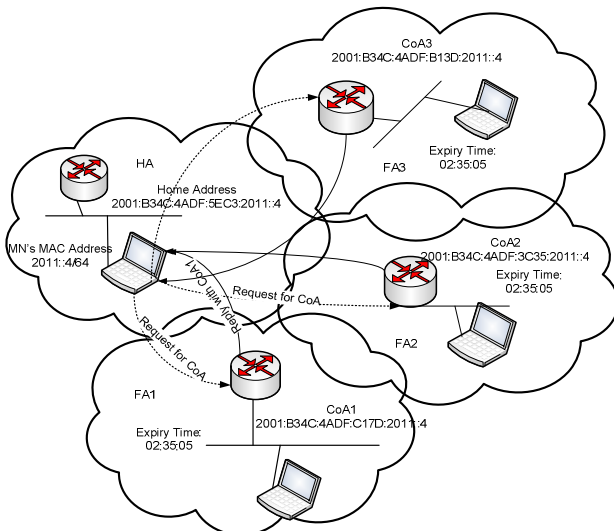


Figure 5 (b). PDAD address configuration scenario

The existing MIPv6 configures the CoA address when it moves to a foreign network in reactive mode. In Fig. 5a, the dotted line shows the request for CoA and another line shows the reply with generated CoA. Here, the MN only requests for one CoA address and the MN configures that address. On the other hand, our mechanism PDAD requests for several CoA to its neighbouring networks in parallel. In Fig. 5b, when the MN detects that it is moving its HA then it sends 3 requests to FA1, FA2 and FA3 in parallel. All 3 requested routers generate 3 CoAs for the said MN and store it in their cache for a time period. If the MN moves to these FAs within the time period it will configure the cached CoA without delay. In this method, the MN takes time only for initial transition and do not need any time for subsequent transition and address configuration as alternative addresses are ready for this MN. The MN does not require configuration time while it moves from FA1 to FA2 or FA2 to FA3. Thus, this mechanism saves the most lengthy configuration time of MIPv6. All requested

routers refresh their cache after the expiration of the time. Each router maintains this cache in its database. However, if the MN moves to a new network it needs to start the same procedure from the beginning.

VII. CONCLUSION AND FUTURE WORK

This paper proposes a parallel DAD (PDAD) mechanism for minimizing handoff delay in advance. Our mechanism addresses the limitations of conventional DAD by requesting for multiple addresses from the neighbor ARs. It works like advanced DAD and maintains a pool of addresses in its cache. The advantage of this mechanism is that it reduces configuration of CoA delay by maintaining a unique address in advance. The configuration of multiple CoAs and DAD procedures may affect the neighboring ARs. However, bandwidth, memory capacity and address consumption of IPv6 are less important issues compared to seamless communication when enhancing the QoS of mobile IP applications. In this paper, we only consider DAD delay where other delays like link layer delay, movement detection delay and binding update delays are not considered. In our future work, we will conduct extensive simulation with OMNET++ considering all types of delay of MIPv6 to improve its QoS.

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